

Percent of Change: Sales Tax, Discounts, Markup & Reverse Problems

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Standard: CCSS.MATH.CONTENT.7.RP.A.3 — Multistep ratio and percent problems.

Part A — Sales Tax and Total Cost

MEDIUM

Problem 1

Radio marked at \$50 with 6% sales tax. Find tax and total cost.

Sales Tax

Tax: $0.06 \times \$50 = \3 .

Total: $\$50 + \$3 = \$53$.

Tax = 3, Total = 53

Problem 2

Bookshelf marked at \$200 with 8% sales tax. Find tax and total cost.

Sales Tax

Tax: $0.08 \times \$200 = \16 .

Total: $\$200 + \$16 = \$216$.

Tax = 16, Total = 216

Part B — Discount and Sale Price

MEDIUM

Problem 3

Sweater list price \$80, 25% discount. Find discount and sale price.

Discount

Discount: $0.25 \times \$80 = \20 .

Sale price: $\$80 - \$20 = \$60$.

Discount = 20, Sale = 60

Problem 4

Jacket list price \$150, 40% discount. Find discount and sale price.

Discount

Discount: $0.40 \times \$150 = \60 .

Sale price: $\$150 - \$60 = \$90$.

Discount = 60, Sale = 90

Problem 5

40 points to 60 points. Increase or decrease? Percent of change?

Percent Increase

Change: $60 - 40 = 20$ (went up → **increase**).

Set up: $\frac{20}{40} = \frac{X}{100}$.

Solve: $X = \frac{20 \times 100}{40} = 50$.

50% Increase

Problem 6

80 to 60. Increase or decrease? Percent of change?

Percent Decrease

Change: $\$80 - \$60 = \$20$ (went down → **decrease**).

Set up: $\frac{20}{80} = \frac{X}{100} \Rightarrow X = 25$.

25% Decrease

Problem 7

25 points to 50 points. Increase or decrease? Percent of change?

Percent Increase

Doubled

Change: $50 - 25 = 25$ (**increase**).

Set up: $\frac{25}{25} = \frac{X}{100} \Rightarrow X = 100$.

Note: When something *doubles*, the change equals the original — that's exactly 100% increase.

100% Increase

Problem 8

125 invitations to 75 invitations. Increase or decrease? Percent of change?

Percent Decrease

Change: $125 - 75 = 50$ (**decrease**).

Set up: $\frac{50}{125} = \frac{X}{100} \Rightarrow X = 40$.

40% Decrease

Part D — Find the Original Amount (Reverse)

MEDIUM

Problem 9

After 20% off, shoes sell for \$40. Find original price.

Reverse Discount

80% remains

After 20% off, sale price = 80% of original.

Equation: $0.80 \times x = 40$.

$$x = \frac{40}{0.80} = \$50.$$

\$50

Problem 10

Including 10% sales tax, hat costs \$44. Find price before tax.

Reverse Tax

110%

Total = 110% of original. Equation: $1.10 \times x = 44$.

$$x = \frac{44}{1.10} = \$40.$$

\$40

▼ CHALLENGING SECTION ▼

Part E — Markup

CHALLENGE

Problem 11

Nike sneakers cost Nike 20 to make; customers pay 150. Find percent of markup.

Markup

Lecture Example

Change: $\$150 - \$20 = \$130$.

Original (cost): \$20.

$$\text{Set up: } \frac{130}{20} = \frac{X}{100} \Rightarrow X = \frac{130 \times 100}{20} = 650.$$

650% Markup

Problem 12

Backpack cost to store: 40. *Selling price* :100. Find percent of markup.

Markup

Change: $\$100 - \$40 = \$60$.

Set up: $\frac{60}{40} = \frac{X}{100} \Rightarrow X = 150$.

150% Markup

Part F — Find the Rate

CHALLENGE

Problem 13

Marked 40, *total cost* 42. Find sales tax rate.

Find Rate

Tax

Tax amount: $\$42 - \$40 = \$2$.

Set up: $\frac{2}{40} = \frac{X}{100} \Rightarrow X = 5$.

5% Sales Tax

Problem 14

List 120, *sale* 90. Find rate of discount.

Find Rate

Discount

Discount amount: $\$120 - \$90 = \$30$.

Set up: $\frac{30}{120} = \frac{X}{100} \Rightarrow X = 25$.

25% Discount

Part G — Multi-Step Word Problems

CHALLENGE

Problem 15

Bicycle \$200. 25% off, then 6% tax on discounted price. Find final total.

Discount → Tax

Step 1 — Discount: $0.25 \times \$200 = \50 . Sale: $\$200 - \$50 = \$150$.

Step 2 — Tax on sale: $0.06 \times \$150 = \9 .

Step 3 — Final: $\$150 + \$9 = \$159$.

\$159

Problem 16

Guitar \$300. 30% off, then 10% tax on discounted price. Find final total.

Discount → Tax

Step 1 — Discount: $0.30 \times \$300 = \90 . Sale: $\$300 - \$90 = \$210$.

Step 2 — Tax: $0.10 \times \$210 = \21 .

Step 3 — Final: $\$210 + \$21 = \$231$.

\$231

Problem 17

Book on sale for 30% off, you pay \$63. Find original price.

Reverse Discount

70% remains

After 30% off, you pay 70% of original.

Equation: $0.70 \times x = 63 \Rightarrow x = \frac{63}{0.70} = \90 .

Check: 30% of 90 = 27; $90 - 27 = \$63$ ✓.

\$90

Problem 18

72 speeding tickets to 36. Increase or decrease? Find percent of change.

Percent Decrease

Halved

Change: $72 - 36 = 36$ (**decrease**).

Set up: $\frac{36}{72} = \frac{X}{100} \Rightarrow X = 50$.

Note: Halving = 50% decrease.

50% Decrease

Problem 19

Store A: 50, 3060, 40% off. Which has the lower sale price, by how much?

Compare

Better Buy

Store A: Discount = $0.30 \times \$50 = \15 . Sale: $\$50 - \$15 = \$35$.

Store B: Discount = $0.40 \times \$60 = \24 . Sale: $\$60 - \$24 = \$36$.

Compare: $\$36 - \$35 = \$1$. Store A is cheaper by \$1.

Lesson: A bigger discount % doesn't always win — the original price matters too.

Store A wins by 1 (Sale = 35)

Problem 20

Jacket \$200. 25% off, then extra 10% off the sale price (coupon). Find final price.

Stacked Discounts

Step 1 — First discount (25%): $0.25 \times \$200 = \50 . Sale: $\$200 - \$50 = \$150$.

Step 2 — Coupon (10% off the new \$150): $0.10 \times \$150 = \15 . New price: $\$150 - \$15 = \$135$.

Trap to avoid: 25% + 10% is NOT the same as 35% off (\$130). The coupon is taken on the already-discounted price, not on the original. Stacked discounts always come out higher than the simple sum.

\$135